

Report Td1-2-3

Maximilien BERNARD

TD 1

Git Clone my branch

```
C:\Users\maxim>git clone -b Maximilien-Bernard "https://github.com/Alex-dmns/esilv-marketpulse-g01-tmhajeur.git" C:\Users\maxim\Documents\ESILV\Python,Git,Linux
Cloning into 'C:\Users\maxim\Documents\ESILV\Python,Git,Linux'...
remote: Enumerating objects: 37, done.
remote: Counting objects: 100% (37/37), done.
remote: Compressing objects: 100% (30/30), done.
remote: Total 37 (delta 6), reused 30 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (37/37), 31.40 KiB | 1.74 MiB/s, done.
Resolving deltas: 100% (6/6), done.

C:\Users\maxim>
```

```
C:\Users\maxim\Documents\ESILV\Python,Git,Linux>git status
On branch Maximilien-Bernard
Your branch is up to date with 'origin/Maximilien-Bernard'.

nothing to commit, working tree clean
```

Part 5

5.1

```
PS C:\Users\maxim\Documents\ESILV\Python,Git,Linux> pwd

Path
----
C:\Users\maxim\Documents\ESILV\Python,Git,Linux
```

5.2

```
PS C:\Users\maxim\Documents\ESILV\Python,Git,Linux> ls

Répertoire : C:\Users\maxim\Documents\ESILV\Python,Git,Linux

Mode                LastWriteTime         Length Name
----                -
d-----          07/10/2026      15:58         .venv
d-----          07/10/2026      15:24        config
d-----          07/10/2026      15:24        data
d-----          07/10/2026      15:24      evidence
d-----          07/10/2026      15:24        labs
d-----          07/10/2026      15:24      readiness
d-----          07/10/2026      15:24        src
-a----          07/10/2026      15:24         199 .gitignore
-a----          07/10/2026      15:24        3126 CONTRIBUTING.md
-a----          07/10/2026      15:24       10747 README.md
-a----          07/10/2026      15:24         135 requirements.txt
-a----          07/10/2026      15:24        1508 TEAM_TEMPLATE.md
```

5.3

```
PS C:\Users\maxim\Documents\ESILV\Python,Git,Linux> cd data
PS C:\Users\maxim\Documents\ESILV\Python,Git,Linux\data> pwd

Path
----
C:\Users\maxim\Documents\ESILV\Python,Git,Linux\data

PS C:\Users\maxim\Documents\ESILV\Python,Git,Linux\data> ls

Répertoire : C:\Users\maxim\Documents\ESILV\Python,Git,Linux\data

Mode                LastWriteTime         Length Name
----                -
d-----          07/10/2026      15:24        sample
```

```
PS C:\Users\maxim\Documents\ESILV\Python,Git,Linux\data> cd sample
PS C:\Users\maxim\Documents\ESILV\Python,Git,Linux\data\sample> pwd
```

```
Path
----
```

```
C:\Users\maxim\Documents\ESILV\Python,Git,Linux\data\sample
```

```
PS C:\Users\maxim\Documents\ESILV\Python,Git,Linux\data\sample> ls
```

```
Répertoire : C:\Users\maxim\Documents\ESILV\Python,Git,Linux\data\sample
```

Mode		LastWriteTime	Length	Name
----		-----	-----	----
-a----	07/10/2026	15:24	7625	bloomberg_reference_expected.json
-a----	07/10/2026	15:24	10281	bloomberg_reference_sample.json
-a----	07/10/2026	15:24	244	instruments.json
-a----	07/10/2026	15:24	2266	prices.csv

```
PS C:\Users\maxim\Documents\ESILV\Python,Git,Linux\data\sample> cd ../../
PS C:\Users\maxim\Documents\ESILV\Python,Git,Linux> pwd
```

```
Path
----
```

```
C:\Users\maxim\Documents\ESILV\Python,Git,Linux
```

Part 6

6.1

```
PS C:\Users\maxim\Documents\ESILV\Python,Git,Linux> cat data/sample/instruments.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}
```

6.2

```
maxim@xoumamilien UCRT64 ~/Documents/ESILV/Python,Git,Linux (Maximilien-Bernard)
$ head data/sample/prices.csv
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
```

```

maxim@xoumamilien UCRT64 ~/Documents/ESILV/Python,Git,Linux (Maximilien-Bernard)
$ grep AAPL data/sample/prices.csv
grep SP500 data/sample/prices.csv
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-03,AAPL,249.60,251.30,248.40,249.80,41000000
2026-09-04,AAPL,251.30,253.60,250.10,252.10,42500000
2026-09-08,AAPL,251.90,253.90,250.70,252.40,44000000
2026-09-09,AAPL,254.00,255.70,252.80,254.20,38000000
2026-09-10,AAPL,254.30,256.60,253.10,255.10,39500000
2026-09-11,AAPL,254.20,256.20,253.00,254.70,41000000
2026-09-14,AAPL,256.60,258.30,255.40,256.80,42500000
2026-09-15,AAPL,257.20,259.50,256.00,258.00,44000000
2026-09-16,AAPL,256.70,258.70,255.50,257.20,38000000
2026-09-17,AAPL,259.20,260.90,258.00,259.40,39500000
2026-09-18,AAPL,259.30,261.60,258.10,260.10,41000000
2026-09-21,AAPL,259.10,261.10,257.90,259.60,42500000
2026-09-22,AAPL,261.10,262.80,259.90,261.30,44000000
2026-09-23,AAPL,261.20,263.50,260.00,262.00,38000000
2026-09-24,AAPL,263.00,265.00,261.80,263.50,39500000
2026-09-25,AAPL,262.60,264.30,261.40,262.80,41000000
2026-09-28,AAPL,263.30,265.60,262.10,264.10,42500000
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-03,SP500,6596.00,6613.00,6584.00,6598.00,0
2026-09-04,SP500,6612.00,6635.00,6600.00,6620.00,0
2026-09-08,SP500,6623.00,6643.00,6611.00,6628.00,0
2026-09-09,SP500,6640.00,6657.00,6628.00,6642.00,0
2026-09-10,SP500,6642.00,6665.00,6630.00,6650.00,0
2026-09-11,SP500,6642.00,6662.00,6630.00,6647.00,0
2026-09-14,SP500,6663.00,6680.00,6651.00,6665.00,0
2026-09-15,SP500,6664.00,6687.00,6652.00,6672.00,0
2026-09-16,SP500,6663.00,6683.00,6651.00,6668.00,0
2026-09-17,SP500,6683.00,6700.00,6671.00,6685.00,0
2026-09-18,SP500,6684.00,6707.00,6672.00,6692.00,0
2026-09-21,SP500,6683.00,6703.00,6671.00,6688.00,0
2026-09-22,SP500,6703.00,6720.00,6691.00,6705.00,0
2026-09-23,SP500,6704.00,6727.00,6692.00,6712.00,0
2026-09-24,SP500,6715.00,6735.00,6703.00,6720.00,0
2026-09-25,SP500,6713.00,6730.00,6701.00,6715.00,0
2026-09-28,SP500,6720.00,6743.00,6708.00,6728.00,0

```

Part 7

7.1

```

maxim@xoumamilien UCRT64 ~/Documents/ESILV/Python,Git,Linux (Maximilien-Bernard)
$ python --version
Python 3.12.10

```

7.2

```
maxim@xoumamilien UCRT64 ~/Documents/ESILV/Python,Git,Linux (Maximilien-Bernard)
$ git --version
git version 2.56.0.windows.2
```

7.3 Run MarketPulse

```
C:\Users\maxim\Documents\ESILV\Python,Git,Linux>python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21

C:\Users\maxim\Documents\ESILV\Python,Git,Linux>|
```

Optional 1

```
maxim@xoumamilien UCRT64 ~/Documents/ESILV/Python,Git,Linux (Maximilien-Bernard)
$ grep -c AAPL data/sample/prices.csv
21
● grep -c SP500 data/sample/prices.csv
21
```

Optional 2

```

maxim@xoumamilien UCRT64 ~/Documents/ESILV/Python,Git,Linux (Maximilien-Bernard)
$ ls -lh
head -n 5 data/sample/prices.csv
tail -n 5 data/sample/prices.csv
total 25K
drwxr-xr-x 1 maxim 197609  0 Oct  7 15:24 config/
-rw-r--r-- 1 maxim 197609 3.1K Oct  7 15:24 CONTRIBUTING.md
drwxr-xr-x 1 maxim 197609  0 Oct  7 15:24 data/
drwxr-xr-x 1 maxim 197609  0 Oct  7 15:24 evidence/
drwxr-xr-x 1 maxim 197609  0 Oct  7 15:24 labs/
drwxr-xr-x 1 maxim 197609  0 Oct  7 15:24 readiness/
-rw-r--r-- 1 maxim 197609 11K Oct  7 15:24 README.md
-rw-r--r-- 1 maxim 197609 135 Oct  7 15:24 requirements.txt
drwxr-xr-x 1 maxim 197609  0 Oct  7 15:24 src/
-rw-r--r-- 1 maxim 197609 1.5K Oct  7 15:24 TEAM_TEMPLATE.md
date,ticker,open,high,low,close,volume
2026-09-01,AAPL,249.20,251.50,248.00,250.00,38000000
2026-09-01,SP500,6592.00,6615.00,6580.00,6600.00,0
2026-09-02,AAPL,250.80,252.80,249.60,251.30,39500000
2026-09-02,SP500,6607.00,6627.00,6595.00,6612.00,0
2026-09-28,SP500,6720.00,6743.00,6708.00,6728.00,0
2026-09-29,AAPL,264.50,266.50,263.30,265.00,44000000
2026-09-29,SP500,6729.00,6749.00,6717.00,6734.00,0
2026-09-30,AAPL,266.00,267.70,264.80,266.20,38000000
2026-09-30,SP500,6740.00,6757.00,6728.00,6742.00,0

```

Head affiche les 5 premières lignes du csv alors que tail affiche les 5 dernières lignes du csv

Optional 3

```

maxim@xoumamilien UCRT64 ~/Documents/ESILV/Python,Git,Linux (Maximilien-Bernard)
$ find . -type f
grep -R "AAPL" .
./.git/config
./.git/description
./.git/HEAD
./.git/hooks/applypatch-msg.sample
./.git/hooks/commit-msg.sample
./.git/hooks/fsmonitor-watchman.sample
./.git/hooks/post-update.sample
./.git/hooks/pre-applypatch.sample
./.git/hooks/pre-commit.sample
./.git/hooks/pre-merge-commit.sample
./.git/hooks/pre-push.sample
./.git/hooks/pre-rebase.sample
./.git/hooks/pre-receive.sample
./.git/hooks/prepare-commit-msg.sample
./.git/hooks/push-to-checkout.sample
./.git/hooks/sendemail-validate.sample
./.git/hooks/update.sample
./.git/index
./.git/info/exclude
./.git/logs/HEAD
./.git/logs/refs/heads/Maximilien-Bernard
./.git/logs/refs/remotes/origin/HEAD
./.git/objects/0c/b0b2f7f33d14c0401f29892946cd34eec5ee01
./.git/objects/0e/58dbfd5799191531b1318ac0e16006a34a93ec
./.git/objects/27/fa13a7d289a2b91f57fdb1290ec08dd8db45a6

```

Optional 4

```

maxim@xoumamilien UCRT64 ~/Documents/ESILV/Python,Git,Linux (Maximilien-Bernard)
$ git status
git log --oneline
On branch Maximilien-Bernard
Your branch is up to date with 'origin/Maximilien-Bernard'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   src/main.py

no changes added to commit (use "git add" and/or "git commit -a")
5207db8 (HEAD -> Maximilien-Bernard, origin/Maximilien-Bernard, origin/Julie-Yerolymos, origin/Alexandre-Damianos) first td session
ba5c9f4 Update README
9fd6c8a Update README
bfb94a6 init projecttemplate

```

TD 2

Part 2

Inspect cat data/sample/instruments.json

```
PS C:\Users\maxim\Documents\ESILV\Python,Git,Linux> cat data/sample/instruments.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
    "ticker": "SP500",
    "name": "S&P 500",
    "currency": "USD",
    "market": "US"
  }
}
PS C:\Users\maxim\Documents\ESILV\Python,Git,Linux>
```

Temporarily inspect the result:

```
print(type(instruments))
print(instruments.keys())
```

```
<class 'dict'>
dict_keys(['instrument', 'benchmark'])
```

Then inspect nested dictionaries:

```
instrument = instruments["instrument"]
benchmark = instruments["benchmark"]

print(instrument["ticker"])
print(instrument["name"])
print(instrument["currency"])

print(benchmark["ticker"])
print(benchmark["name"])
```

```
SP500, US
AAPL
Apple Inc.
USD
SP500
S&P 500
```

Part 3

Temporarily inspect:

```
prices = load_prices()
```

```

print(type(prices))
print(type(prices[0]))
print(prices[0])

```

```

<class 'list'>
<class 'dict'>
{'date': '2026-09-01', 'ticker': 'AAPL', 'open': '249.20', 'high': '251.50', 'low': '248.00', 'close': '250.00', 'volume': '38000000'}

```

Inspect:

```

print(prices[0]["close"])
print(type(prices[0]["close"]))

```

```

250.00
<class 'str'>

```

Convert when a numerical value is needed:

```

print(f"{instrument['ticker']}: {len(instrument_prices)}")
print(f"{benchmark['ticker']}: {len(benchmark_prices)}")

instrument_close = float(instrument_latest["close"])
benchmark_close = float(benchmark_latest["close"])

```

Part 4

```

instrument_prices = filter_prices(prices, instrument["ticker"])
benchmark_prices = filter_prices(prices, benchmark["ticker"])

```

Part 5

```

def get_first_close(prices):
    return float(prices[0]["close"])

def get_last_close(prices):
    return float(prices[-1]["close"])

```

```

First close : 250.00 USD
Last close  : 266.20 USD

```

Part 6

```
def display_market_summary(asset, prices, show_currency=True):
    currency = f" {asset['currency']}" if show_currency else ""

    print(f"{asset['ticker']} - {asset['name']}")
    print(f"Observations : {len(prices)}")
    print(f"First close : {get_first_close(prices):.2f}{currency}")
    print(f>Last close : {get_last_close(prices):.2f}{currency}")
```

```
Instrument
AAPL - Apple Inc.
Observations : 21
First close : 250.00 USD
Last close : 266.20 USD

Benchmark
SP500 - S&P 500
Observations : 21
First close : 6600.00
Last close : 6742.00
```

Part 8

```
PS C:\Users\maxim\Documents\ESILV\Python,Git,Linux> C:\Users\maxim\AppData\Local\Programs\Python\Python312\python.exe c:/Users/maxim/D
ocuments/ESILV/Python,Git,Linux/src/main.py
=== MarketPulse ===

Market configuration
Period : 1 month
Interval : Daily

Instrument
AAPL - Apple Inc.
Observations : 21
First close : 250.00 USD
Last close : 266.20 USD

Benchmark
SP500 - S&P 500
Observations : 21
First close : 6600.00
Last close : 6742.00
PS C:\Users\maxim\Documents\ESILV\Python,Git,Linux> []
```

FOR Part 9-10, full code pushed in GitHub Branch